30 July 1974
SUPERSEDING
MIL-G-45937
24 April 1973

MILITARY SPECIFICATION SHEET

GUN, AUTOMATIC, 20MM: M168

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the gun described herein shall consist of this document and the issue in effect of specification MIL-G-45500.

1. SCOPE

1.1 This specification sheet covers one type of multibarrel, electrically fired, 20 millimeter (mm), automatic gun. The gun utilizes an ammunition feeder, recoil adapters and is power driven through its cycle of operation by an electric drive.

2. APPLICABLE DOCUMENTS

SPECIFICATIONS

Military

MIL-G-45500 - Gun, Automatic, 20MM: General Specification For

DRAWINGS

U.S. Army Weapons Command

F8436805 - Gun, Automatic, 20MM; M168

PUBLICATIONS

U.S. Army Weapons Command

P8436805 - Packaging Data Sheet for Gun, Automatic, 20MM: M168.

IEL8436805 - Index of Inspection Equipment List for Gun, Automatic, 20MM: M168

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 20261O-22T03:15Z
Check the source to verify that this is the current version before use.

3.1 Material, construction and design. Guns and parts shall conform to the material, construction and design requirements specified herein and on Drawing F8436805.

3.1.1 Case chute assembly. The case chute assembly shall be securely fastened to the housing assembly.

3.1.2 Functioning.

TABLE I

Performance Requirements

Gun Torque (Max)	Acceleration Time (Max)	Stead y State Rate of Fire	Stopping Time (Max)
500 pound-inch at minimum steady state rate of fire	0.35 seconds (from start of gun rotation to 2700 rounds per minute)	2700 to 3000 rounds per minute	0.35 seconds (from minimum steady state rate of fire to complete stop of gun)

3.2 Targeting and accuracy.

3.2.1 When using the M61A1 (11010336) muzzle clamp, guns shall place 80 percent of a 100 round burst completely within an eight (8) inch diameter circle at a range of 1,000 inches for accuracy, and the center of the 80 percent group shall be within 2.5 inches elevation or depression and 4.75 inches azimuth right or left of the true boresight point for targeting.

3.2.2 When using the M168 (8436804) muzzle clamp, guns shall place 80 percent of a 100 round burst completely within a rectangle, eight (8) inches in heights, and eighteen (18 inches in width, at a range of 1,000 inches for accuracy, and the geometric center of the 80 percent group shall be not more than 4 ± 3 inches elevation and 0 ± 3 inches right or left azimuth of the true boresight point for targeting.

3.3 Endurance. Guns shall fire 30,000 rounds for endurance. A series is 1,000 rounds using five (5) bursts of 200 rounds each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:15Z
Check the source to verify that this is the current version before use.

TABLE II

Schedule of Replacement Parts

Part Nomenclature	Part No.	Gun Rounds ± 1,000 Rounds
Pin, spring (for block, locking)	11698266	15,000
Shaft assembly, bolt		15,000
Insulator, pin, cam, firing	11010160	15,000
Insulator assembly, firing pin	11010159	15,000
Pin, cam, firing	7268304	15,000
Pin, firing	7268303	15,000
Spring, firing pin	11698237	15,000
Stop, contact assembly	7268730	15,000

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality conformance inspection.

4.1.1 Examination. Guns shall be visually and manually examined in accordance with the following:

TABLE III

Part or assembly	Reference paragraph
Link and Case chute assembly	3.2

4.1.2 Inspection equipment. The measuring and testing equipment shall be as specified on the Index of Inspection Equipment Lists IEL8436805.

4.2 Test methods.

4.2.1 Function firing test procedure. The gun shall be function fired at a steady state rate of 2,700 to 3,000 rounds per minute using the power supply with control equipment, feeder and calibrated drive specified in MIL-G-45500. To calculate gun torque at minimum steady rate of fire, the steady state rate of fire, and the mean differential pressure across the drive shall be calculated using the last 50 percent of the burst excluding stopping time. The calculated rate of fire shall be 2,700 to 3,000 rounds per minute.

4.2.2 Interchangeability test. The following groups of non-mating parts shall be utilized in the interchangeability test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:15Z
Check the source to verify that this is the current version before use.

GROUPS OF NON-MATING PARTS

GROUP I

Adapter, recoil assembly (8436802)
Barrel (7790801)
Rotor assembly, rear (8437123)
Housing assembly, rear (11010476)
Insulator, pin, cam, firing (11010160)
Pin, firing (7268303)
Stop, contact assembly (7268730)
Track. rotor center (7268703)
Track; rotor front (7268711)
Track, rotor, removable (7268552)
Block, lock (7269955)

GROUP II

Bar, guide (7269918)
Cam, unlocking, front (11698265)
Clamp, center-barrel, assembly (7790616)
Insulator assembly, firing pin (11010159)
Pin, cam, firing (7268304)
Pin, cover, lock (11010132)
Plate, end (8436990)
Retainer assembly (7269886)
Shaft assembly, bolt (7790708)
Spacer, removable track (7268715)
Spring, firing pin (11698237)
Spring, helical, torsion (7791606)

GROUP III

Body, breech bolt (11010158)
Can, lock (7790681)
Cam, unlocking, rear (11698264)
Chute, adapter; assembly (8436803)
Clamp assembly, muzzle (8436804)
Clamp, coupling (7269760)
Contact assembly, firing (11010186)
Cover, housing assembly (7268556)
Pin assembly, indexing (7790768)
Rotor assembly (Remaining parts)
Sector, clearing cam assembly (7268791)

5. PREPARATION FOR DELIVERY

Packaging Data Sheet P8436805 applies.

Custodians:
Army - WC
Navy - OS
Air Force - 84

Review activities:
Navy - OS
Air Force - 84

User activity:
Navy - MC

Preparing activity:
Army-WC

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-G-45500/3; Gun, Automatic, 20MM: M168

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:15Z
Check the source to verify that this is the current version before use.